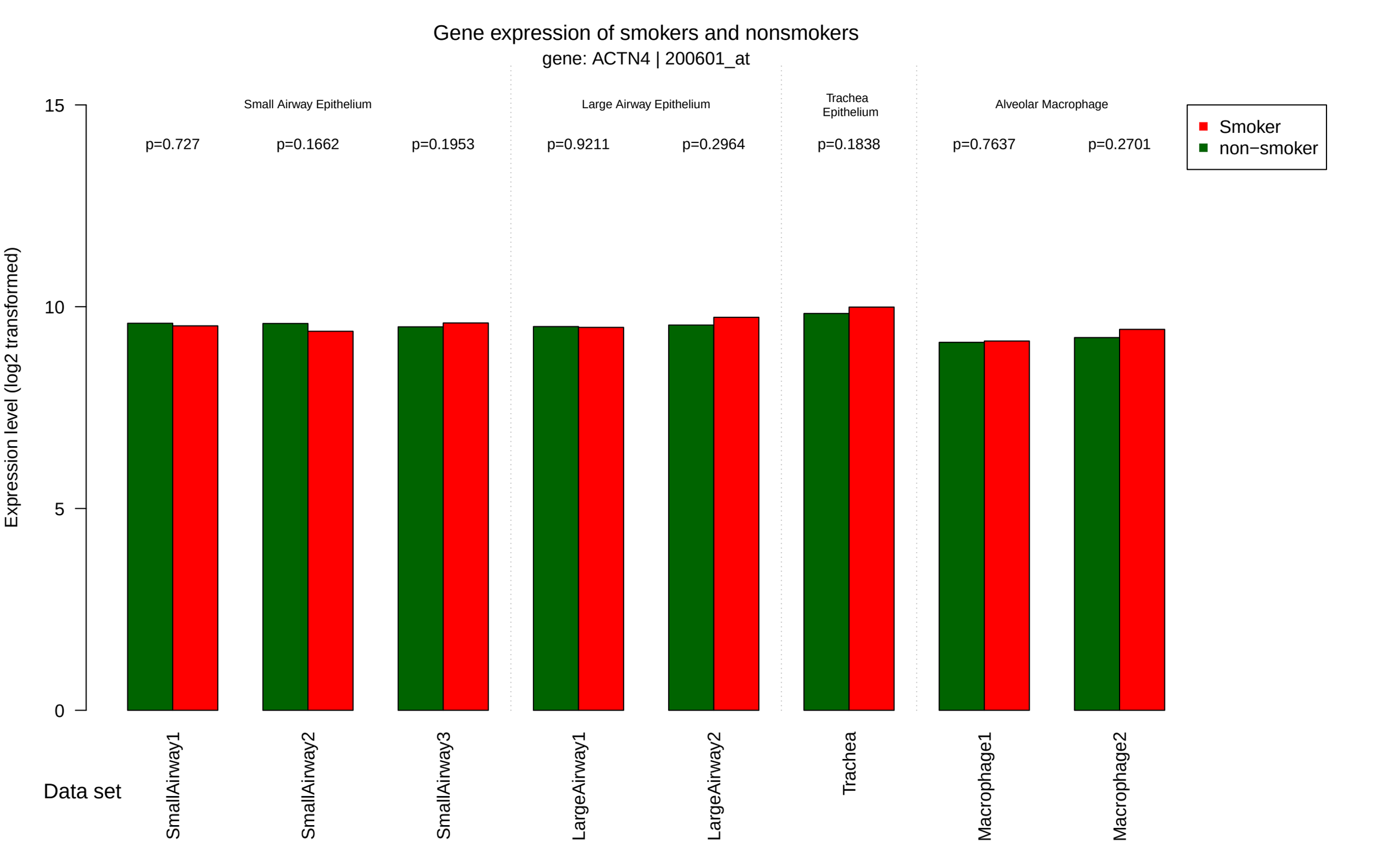
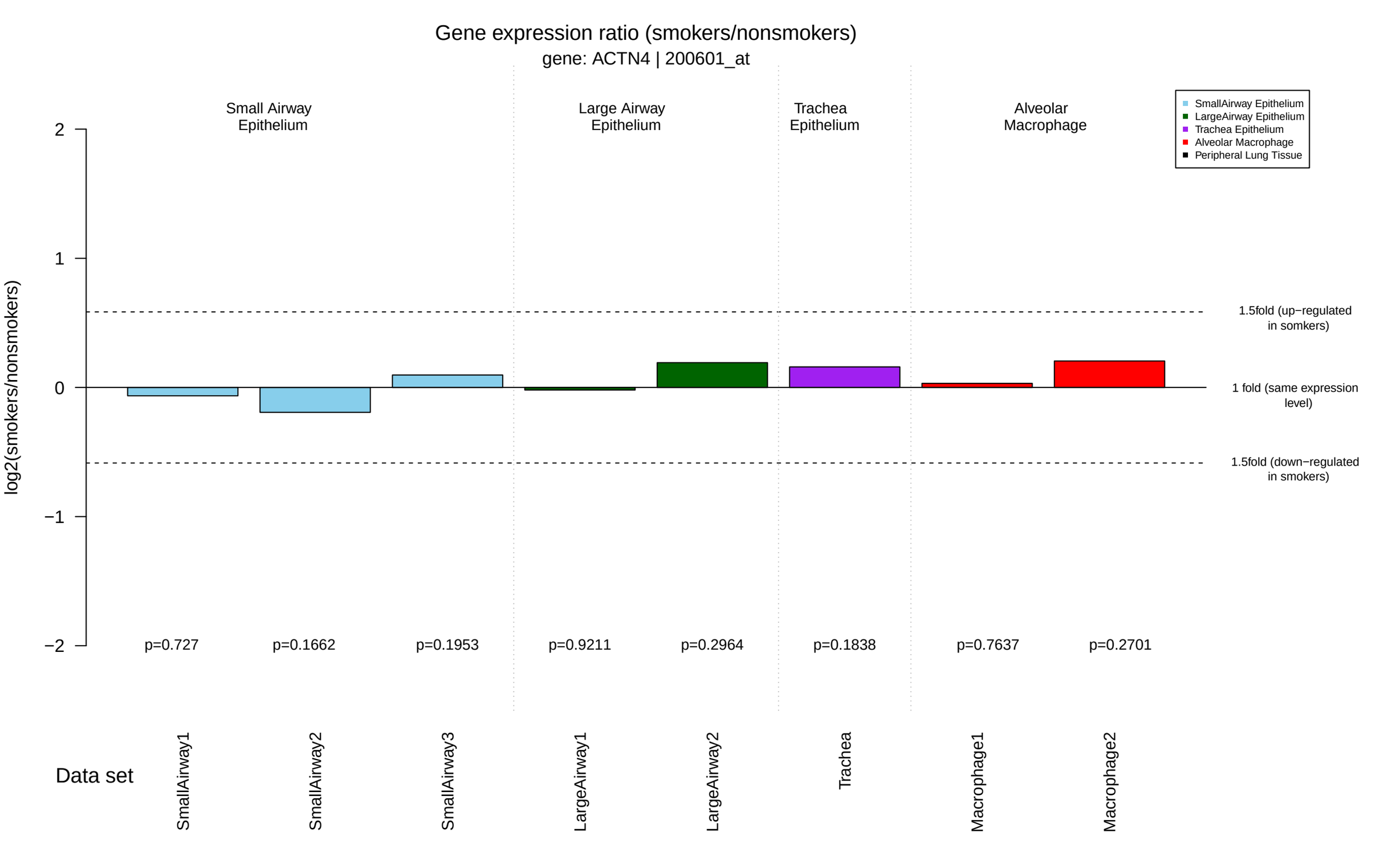
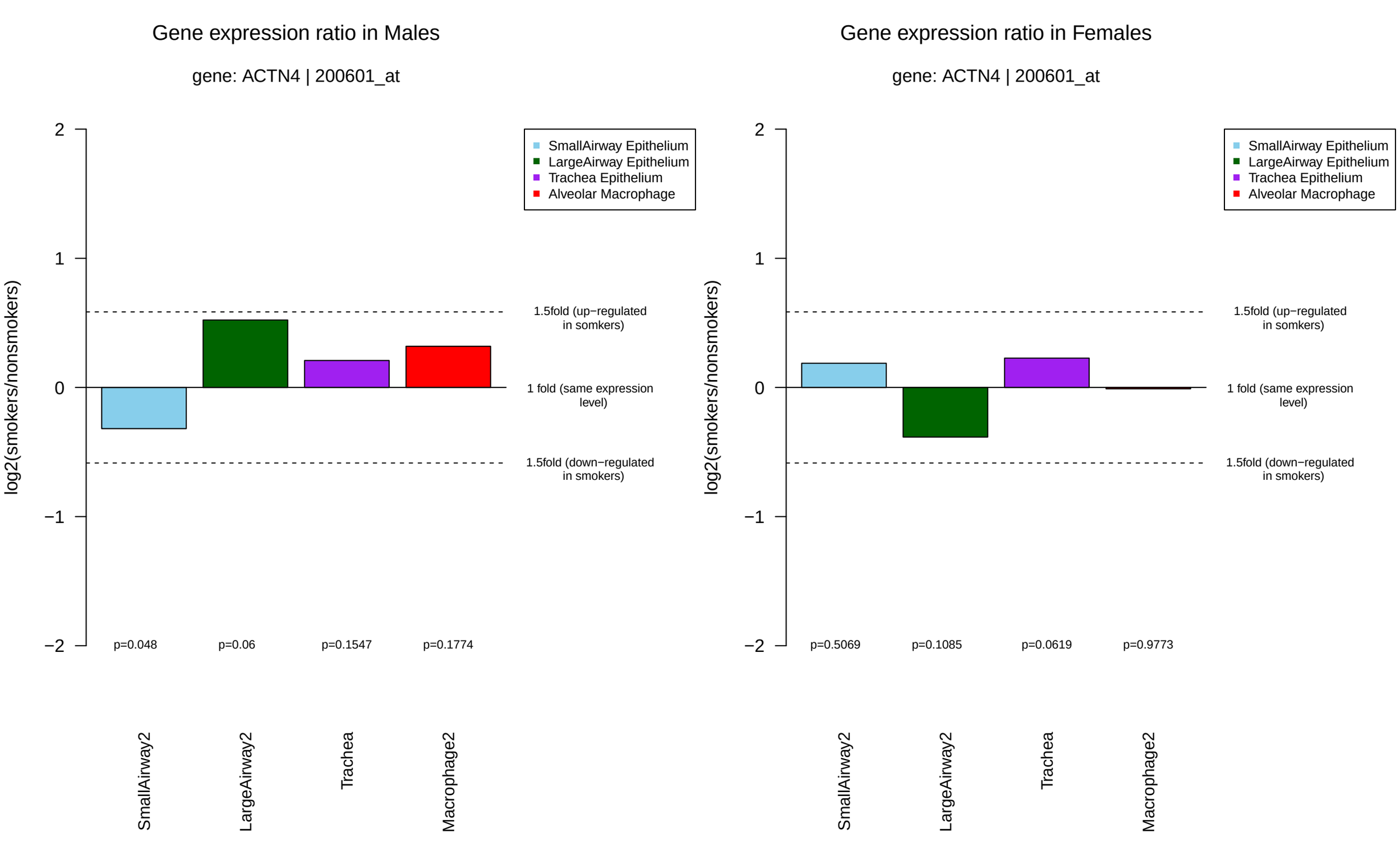
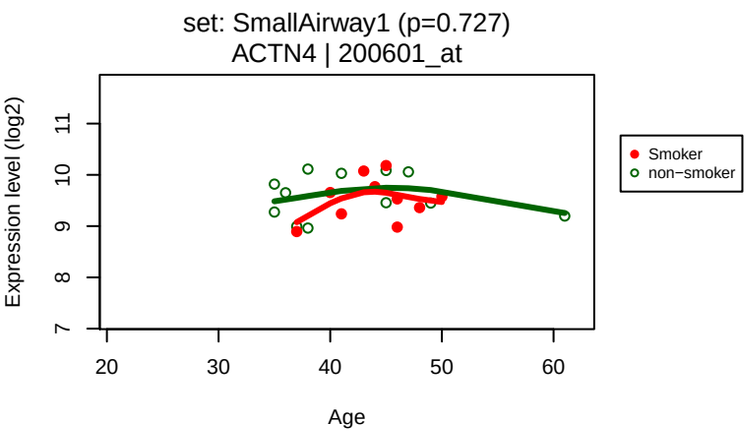
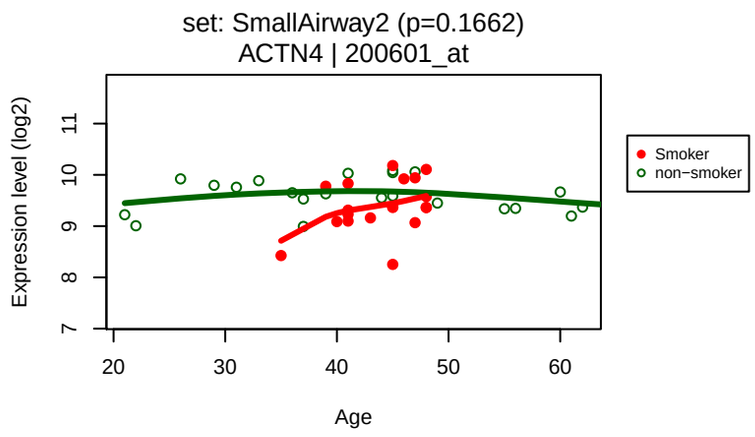
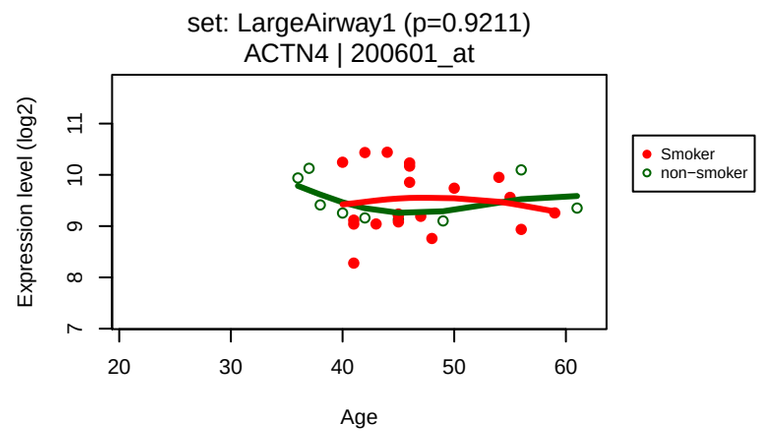

Gene expression of smokers and nonsmokers
gene: ACTN4 | 200601_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.727
p=0.1662
p=0.1953
p=0.9211
p=0.2964
p=0.1838
p=0.7637
p=0.2701
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

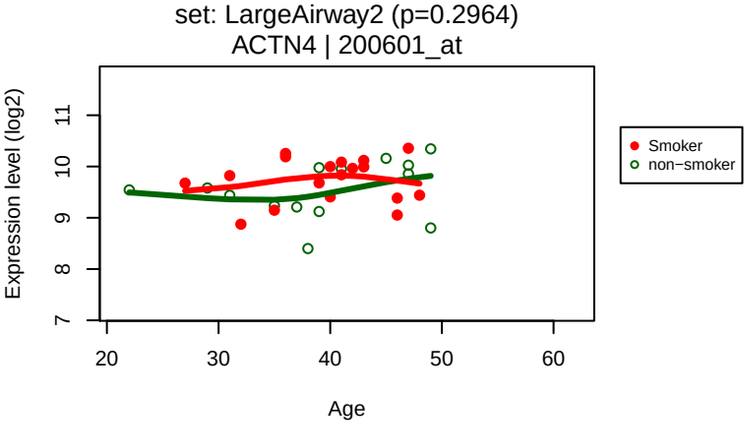

Gene expression ratio (smokers/nonsmokers)
gene: ACTN4 | 200601_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.727
p=0.1662
p=0.1953
p=0.9211
p=0.2964
p=0.1838
p=0.7637
p=0.2701
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

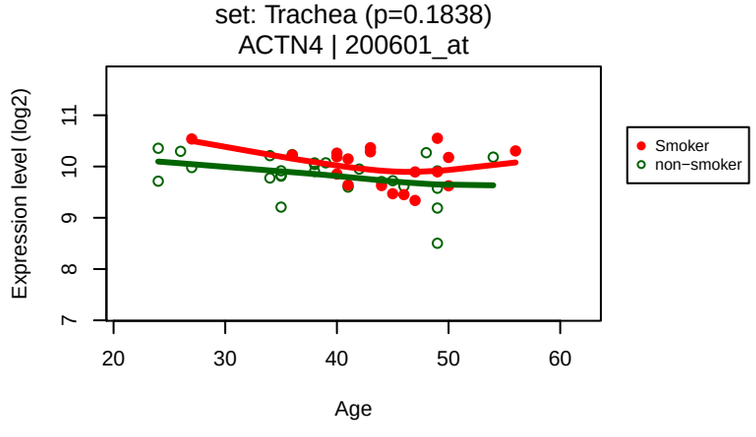

Gene expression ratio in Males
gene: ACTN4 | 200601_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.048
p=0.06
p=0.1547
p=0.1774
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN4 | 200601_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5069
p=0.1085
p=0.0619
p=0.9773
SmallAirway2
LargeAirway2
Trachea
Macrophage2

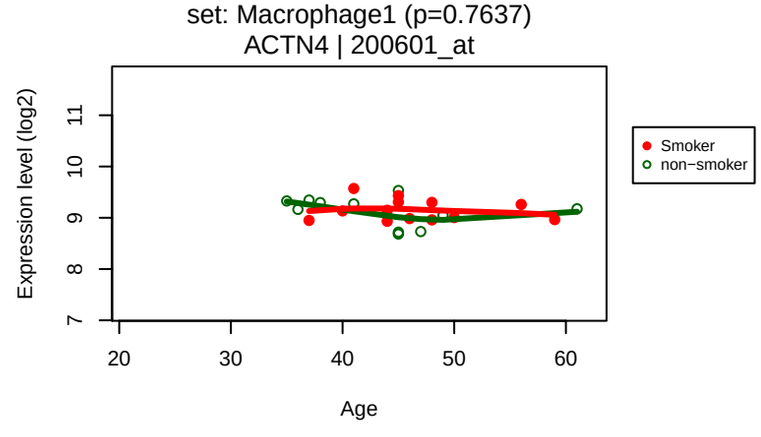

set: SmallAirway1 (p=0.727)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker

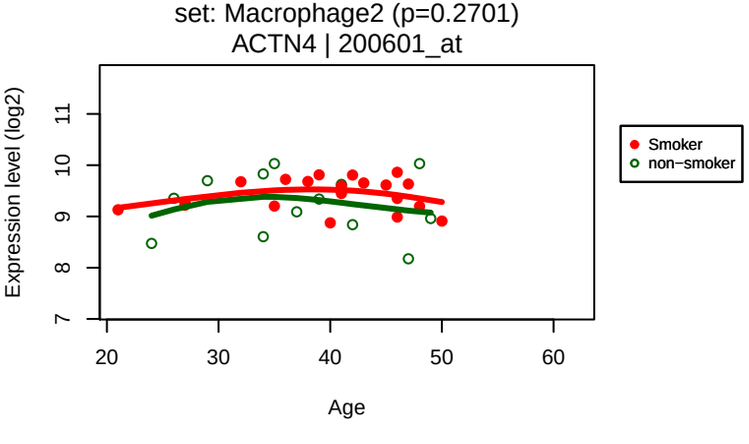

set: SmallAirway2 (p=0.1662)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker

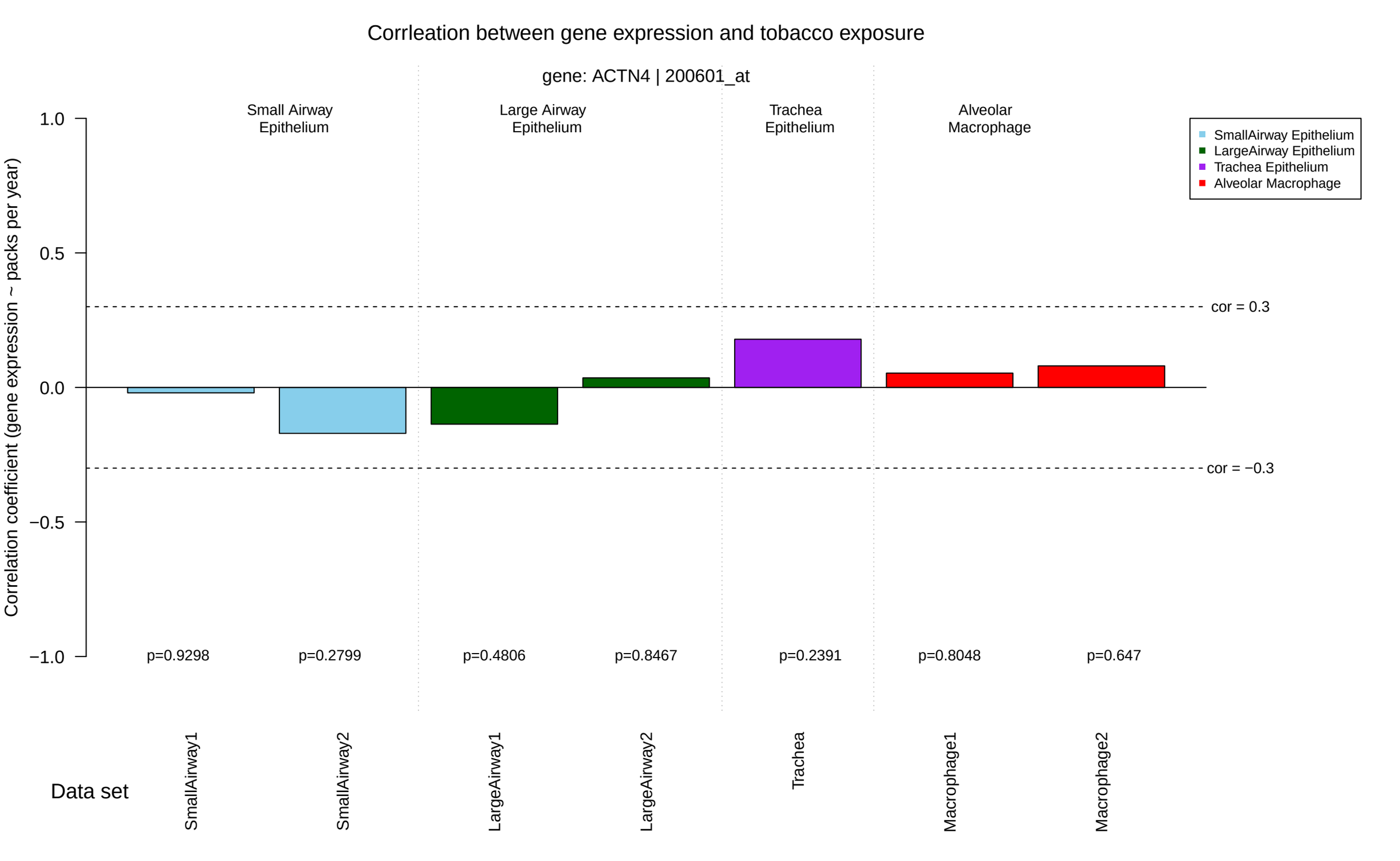

set: LargeAirway1 (p=0.9211)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2964)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1838)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7637)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2701)
ACTN4 | 200601_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN4 | 200601_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9298
p=0.2799
p=0.4806
p=0.8467
p=0.2391
p=0.8048
p=0.647
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2